

- Q.29 Write a short note on DC drive Control. (CO4)
- Q.30 Explain working of single phase full wave controlled rectifier. (CO2)
- Q.31 Draw and explain the circuit diagram of on-line UPS. (CO5)
- Q.32 Write short note on snubber circuit. (CO1)
- Q.33 Illustrate the parallel operation of SCRs with the help of diagram. (CO1)
- Q.34 Write the advantages of HVDC transmission over HVAC transmission. (CO5)
- Q.35 Define inverter. Give comparison of CSI and VSI. (CO4)

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What are different types of commutation methods of SCR? Explain any two. (CO1)
- Q.37 What do you mean by chopper? Explain class B chopper with circuit diagram. (CO3)
- Q.38 Write short note on any two:
- a) Inverter controlled drives. (CO5)
  - b) Dual converter (CO3)
  - c) V-I characteristics of SCR (CO1)

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#### 5th Sem / Mechatronics Sub : Power Electronics

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 LASCR stands for \_\_\_\_\_. (CO1)
- a) Light activated SCR
  - b) Light Anti SCR
  - c) Linear activated SCR
  - d) None of the above
- Q.2 Holding current is \_\_\_\_\_ than latching current. (CO1)
- a) More
  - b) Less
  - c) Equal
  - d) All of the above
- Q.3 The methods of turning ON the thyristor is/are \_\_\_\_\_. (CO1)
- a) Gate triggering
  - b) Light triggering
  - c)  $dV/dt$  triggering
  - d) All of the above
- Q.4 Choppers are \_\_\_\_\_. (CO3)
- a) DC to DC converters
  - b) AC to DC converters
  - c) AC to AC Converters
  - d) DC to AC Converters

- Q.5 The values of duty cycle (a) lies between (CO4)  
 a)  $0 < a < 1$  b)  $0 > a > 10$   
 c)  $0 \leq a \leq 1$  d)  $1 < a < 100$
- Q.6 A dual converter can be operated in \_\_\_\_\_ quadrant without any change over switch (CO4)  
 a) 4 b) 3  
 c) 2 d) None
- Q.7 \_\_\_\_\_ is used in aircraft power supplies provide variable frequency power. (CO4)  
 a) Dual converter b) Cyclo - converter  
 c) Chopper converter d) Rectifier converter
- Q.8 SMPS gives \_\_\_\_\_ output (CO5)  
 a) Pulsating DC b) Voltage sweep  
 c) AC d) DC
- Q.9 SCR can be used as a \_\_\_\_\_ switch. (CO4)  
 a) Bilateral b) Unilateral  
 c) Both d) None of the above
- Q.10 \_\_\_\_\_ are used to build an electric drive? (CO4)  
 a) Source b) Motor  
 c) Control Unit d) All of the above

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Draw the symbol of DIAC. (CO1)

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- Q.12 Expand SCR. (CO1)
- Q.13 Expand MOSFET. (CO1)
- Q.14 Define natural commutation. (CO2)
- Q.15 Define Chopper. (CO3)
- Q.16 What is the use of FWD? (CO2)
- Q.17 Define the term duty cycle. (CO2)
- Q.18 DIAC is used to \_\_\_\_\_. (CO1)
- Q.19 What is the use of electric drive. (CO5)
- Q.20 An SCR is a \_\_\_\_\_ triggered device. (Voltage / Current) (CO4)

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working operation of SCR. (CO1)
- Q.22 Explain the V-I characteristics of DIAC. (CO1)
- Q.23 Write short notes on selection of heat sink. (CO1)
- Q.24 Explain the application of TRIAC as illumination control. (CO2)
- Q.25 Explain load commutation method of SCR with circuit diagram. (CO2)
- Q.26 Explain the working of single phase full wave centre tap controlled rectifier. (CO2)
- Q.27 What do you mean by cyclo-converter? Write its applications. (CO4)
- Q.28 Explain the use of dual converter in speed reversal of a DC motor. (CO3)

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